CS341 · COMPUTER GRAPHICS

Calculations Solutions

Practice on every calculation type

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01 Solutions Q1–Q3

Q1

a) $AR = \dfrac{\frac{768}{9}}{\frac{1024}{12}} = \dfrac{85.33}{85.33} = 1$

b) $AR = \dfrac{\frac{600}{8}}{\frac{800}{10}} = \dfrac{75}{80} = 0.9375$

Q2

x: $\dfrac{1280}{32} = 40$ pixels per cm

y: $\dfrac{960}{24} = 40$ pixels per cm

$AR = \dfrac{40}{40} = 1$

Q3

pixels = 1024 × 768 = 786,432

a) 786,432 bits ÷ 8 = 98,304 bytes

b) 786,432 × 8 bits ÷ 8 = 786,432 bytes (768 KB)

c) 786,432 × 24 bits ÷ 8 = 2,359,296 bytes (2.25 MB)

02 Solution Q4: vectors

a) $v = Q - P = (7 - 2,\ 1 - 5) = (5, -4)$ $|v| = \sqrt{(25 + 16)} = \sqrt{41} \approx 6.4$

b) $|a| = \sqrt{(1 + 4 + 4)} = 3$ $\hat{a} = (\frac{1}{3}, \frac{2}{3}, \frac{2}{3})$

c) $a \cdot b = 1(2) + 2(-1) + 2(0) = 0$ → perpendicular

d) $a \times b = (a_y \cdot b_z - a_z \cdot b_y,\ a_z \cdot b_x - a_x \cdot b_z,\ a_x \cdot b_y - a_y \cdot b_x)$

$= (2 \cdot 0 - 2 \cdot (-1),\ 2 \cdot 2 - 1 \cdot 0,\ 1 \cdot (-1) - 2 \cdot 2)$

$= (2, 4, -5)$

check: $(2, 4, -5) \cdot a = 2 + 8 - 10 = 0$ ✓

e) $u \cdot w = -1$ $|u| = 1$ $|w| = \sqrt{2}$

$\cos\theta = -\frac{1}{\sqrt{2}}$ → $\theta = 135°$

f) $c_\perp = (-c_y, c_x) = (2, 3)$

03 Solution Q5: points and lines

Combination	Sum of coefficients	Valid point?
0.5P + 0.5R	1	**Yes (midpoint)**
2P − R	1	**Yes**
P + R	2	**No**
0.2P + 0.3R	0.5	**No**

```
b) P = (1, 2) + 0.75(8, 4) = (1 + 6, 2 + 3) = (7, 5)
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c) b = B − C = (3, 4)          L(t) = (2, 1) + t(3, 4)
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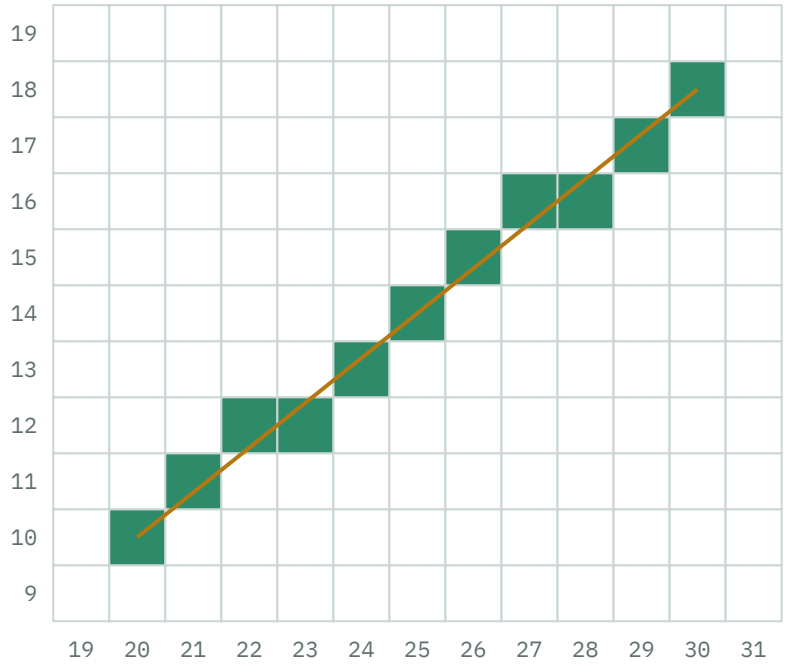
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   n = b⊥ = (−4, 3)            −4(x − 2) + 3(y − 1) = 0
```

04 Solutions Q6–Q8: DDA

Q6: (20,10) → (30,18)

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dx = 10, dy = 8, steps = 10, xInc = 1, yInc = 0.8
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k	x	y	round(x)	round(y)
0	20	10	20	10
1	21	10.8	21	11
2	22	11.6	22	12
3	23	12.4	23	12
4	24	13.2	24	13
5	25	14	25	14
6	26	14.8	26	15
7	27	15.6	27	16
8	28	16.4	28	16
9	29	17.2	29	17
10	30	18	30	18

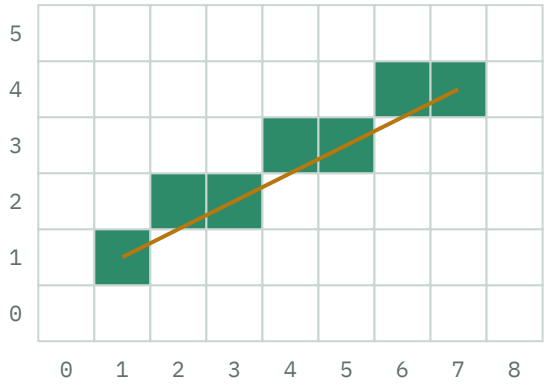


Q7: (1,1) → (7,4)

$$dx = 6,\ dy = 3,\ steps = 6,\ xInc = 1,\ yInc = \frac{3}{6} = 0.5$$

$$round(a) = int(a + 0.5) \quad \rightarrow \quad 1.5 \rightarrow 2, \quad 2.5 \rightarrow 3, \quad 3.5 \rightarrow 4$$

k	x	y	round(x)	round(y)
0	1	1	1	1
1	2	1.5	2	2
2	3	2	3	2
3	4	2.5	4	3
4	5	3	5	3
5	6	3.5	6	4
6	7	4	7	4



Q8: (0,0) → (3,6)

$$dx = 3,\ dy = 6,\ m = 2 > 1,\ steps = 6,\ xInc = \frac{3}{6} = 0.5,\ yInc = 1$$

k	x	y	round(x)	round(y)
0	0	0	0	0
1	0.5	1	1	1
2	1	2	1	2
3	1.5	3	2	3
4	2	4	2	4
5	2.5	5	3	5
6	3	6	3	6

